Curvature and Torsion in the Continuum Theory of Lattice Defects

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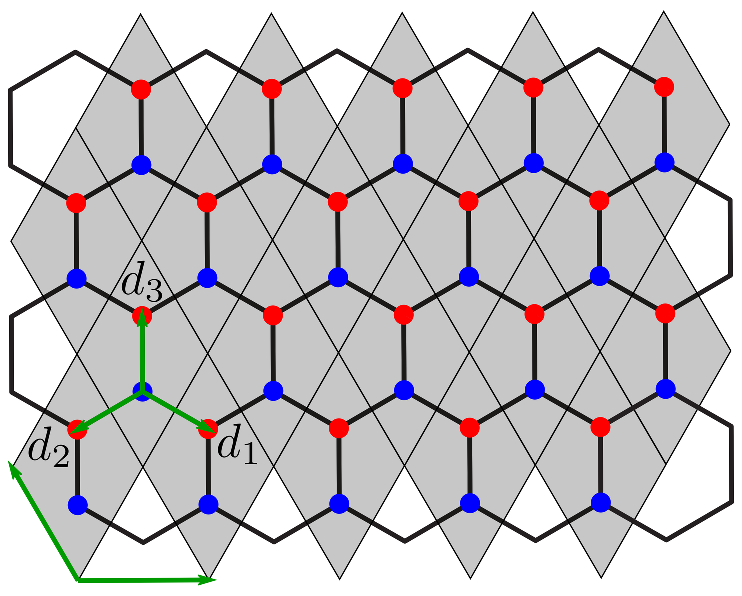
Offen im Denken

Continuum Theory - Graphene



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^{3} \left(a^\dagger_{\vec{r}} b_{\vec{r}+\vec{d}_j} + b^\dagger_{\vec{r}+\vec{d}_j} a_{\vec{r}} \right)$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \boldsymbol{h}(\vec{k}) \vec{\psi}(\vec{k})$$

$$\boldsymbol{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^{3} e^{i\vec{k}\cdot\vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \vec{\delta k} \quad \Longrightarrow \quad \boldsymbol{h}(\vec{k}) \approx -\frac{3}{2} \vec{\delta k} \cdot \vec{\boldsymbol{\sigma}}$$

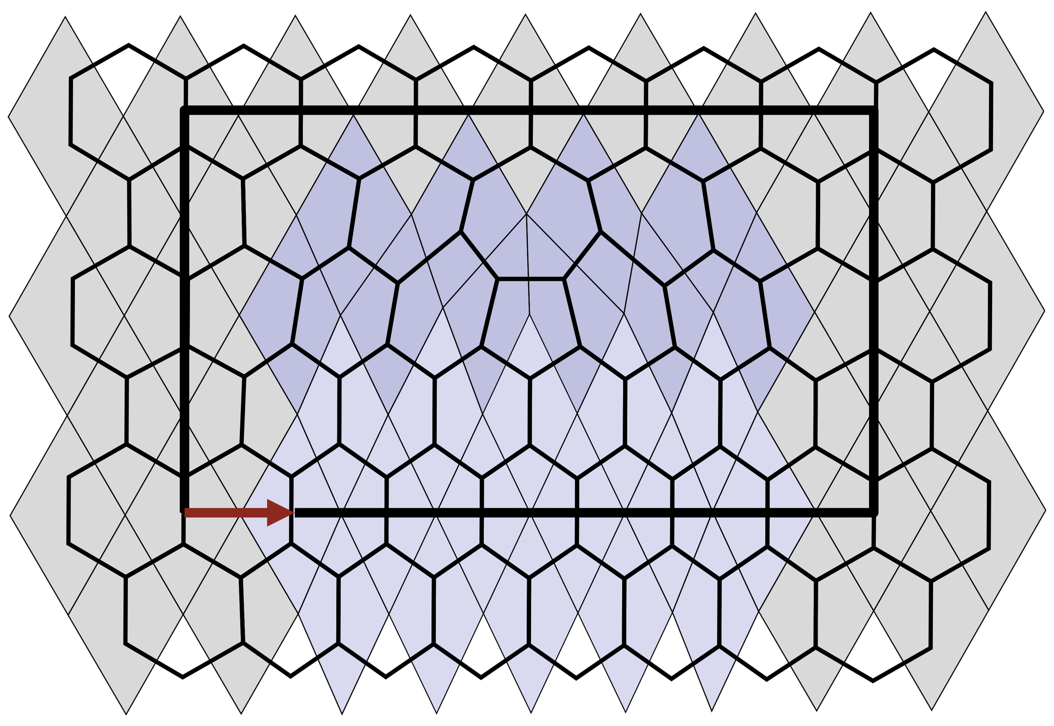
Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i\, c\, \boldsymbol{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \qquad \rho = \vec{\phi}^\dagger \vec{\phi}, \qquad \vec{j} = -c \vec{\phi}^\dagger \vec{\boldsymbol{\sigma}} \vec{\phi}$$

$$\Updownarrow$$

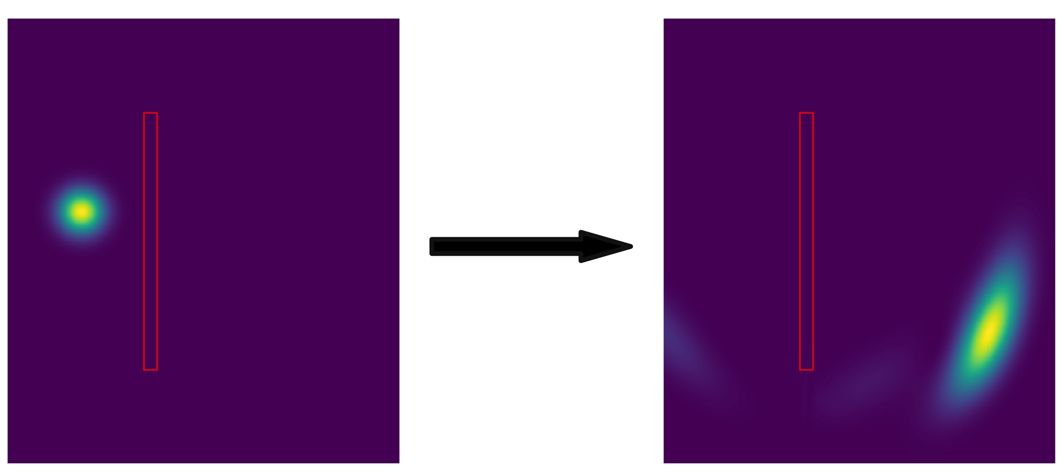
$$i\gamma^\mu \partial_\mu \phi = 0, \quad \gamma^\mu = \sigma^\mu \sigma^3, \quad \mu \in \{0, 1, 2\}$$

Burgers vector in 5-7 defects



One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = \sqrt{3}\, \vec{u}_x$

TODO:
Numerical implementation of
the tight binding hamiltonian
- vector matrix system $\rightarrow$ ODE
- unitary timeevolution
- example plot of density before and after scattering

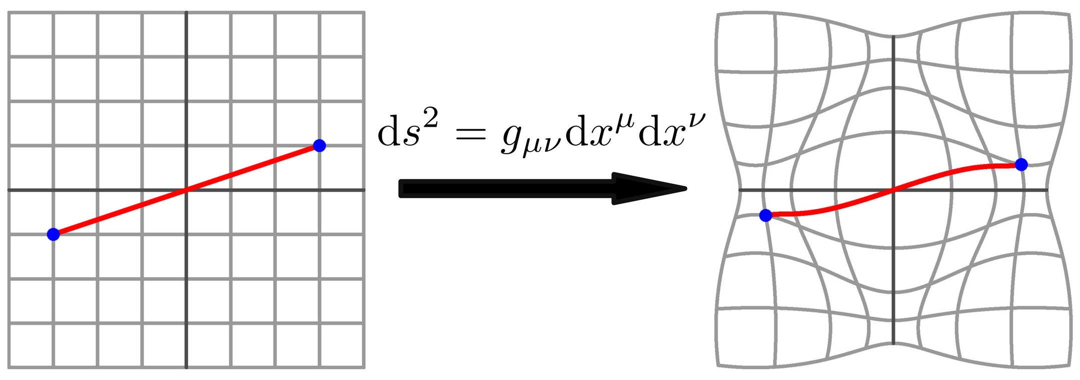


Riemannian Geometry

Riemannian Manifold $\mathcal{M}$ equipped with metric $g_{\mu\nu}$ and connection $\Gamma^\lambda{}_{\mu\nu}$.

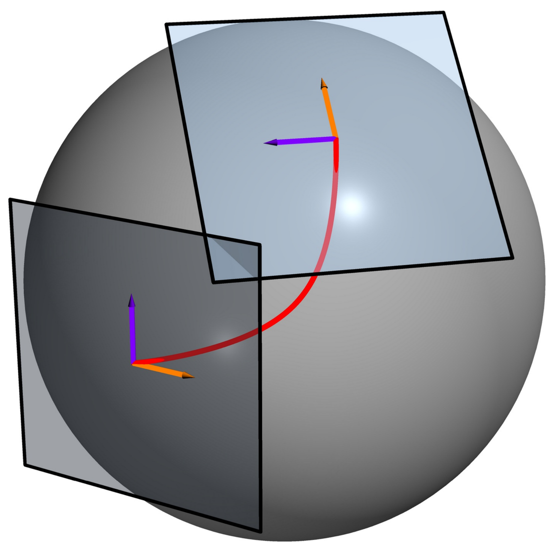
Metric

Defines distances and angles.



Connection

Connectes nearby **tangent spaces**, which allows the definition of a derivative on **vector fields** which take on values in the tangent space.



The unique torsion-free, metric compatible connection is called the **Levi-Civita connection**:

$$\left\{ {}^{\lambda}_{\mu\nu} \right\} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu} \right)$$

Covariant derivative

Takes into account the structure of the manifold and transforms like a tensor.

$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu{}_{\mu\lambda} v^\lambda$$

Autoparallels

A curve whose tangent vector is parallel-transported along itself.

$$u^\mu \nabla_\mu u^\alpha = 0$$

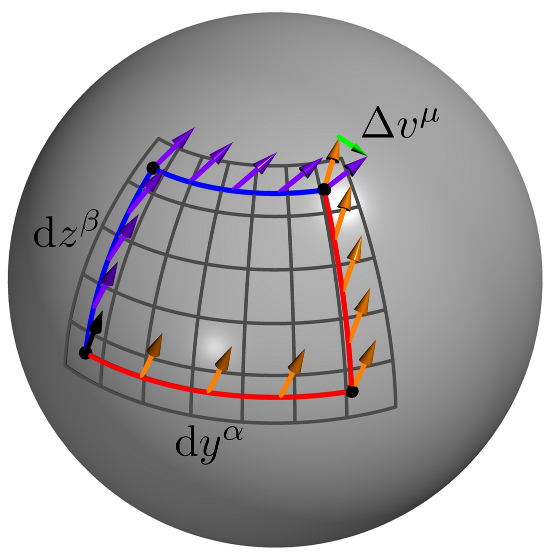
Curvature

The Riemann curvature tensor describes how parallel transport of vectors depends on the path.

$$\Delta v^\mu = R_{\alpha\beta}{}^\mu{}_\nu\, v^\nu\, \mathrm{d}y^\alpha\, \mathrm{d}z^\beta$$

It can be expressed in terms of the connection:

$$R_{\alpha\beta}{}^\mu{}_\nu = \partial_\alpha \Gamma^\mu{}_{\beta\nu} - \partial_\beta \Gamma^\mu{}_{\alpha\nu} + \Gamma^\mu{}_{\alpha\lambda} \Gamma^\lambda{}_{\beta\nu} - \Gamma^\mu{}_{\beta\lambda} \Gamma^\lambda{}_{\alpha\nu}$$



Metric $\leftrightarrow$ torsion mapping in 2D

In 2D we propose a mapping between a **classical** Manifold $\mathcal{M}_g(g_{\mu\nu}, \{{}^\lambda_{\mu\nu}\})$ and **purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k{}_{ij}\})$.
In 2D we can always choose coordinates in which the metric is $g_{\mu\nu} = \delta_{\mu\nu}\, e^{2\phi(x,y)}$.
Also the torsion in 2D can be written using a single covector $T^k{}_{ij} = b^k \varepsilon_{ij}$
We demand equal **curvature** of both manifolds:

$$R\,[\mathcal{M}_g] = R\,[\mathcal{M}_T]\,,$$

This leads to the following solution for the torsion vector b^i as a function of the metric ϕ and an arbitrary scalar field χ:

$$b^i = \varepsilon_{ik}\, \partial_k \phi + \partial_i \chi$$

For $\chi = 0$ the **autoparallels** of both manifolds are equivalent up to a reparameterization. It can be shown, that the χ field corresponds to local roatations of the frame field.

Conclusion

Except for distances and the problem of different tangent spaces a physical mapping between torsion and metric is possible.

Torsion

In standard GR the torsion is assumed to vanish. It is defined as the antisymmetric part of the connection:

$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\mu}$$

Any general affine connection can be decomposed into a symmetric part, the contorsion and the disformation:

$$\Gamma^\alpha{}_{\mu\nu} = \{{}^\alpha{}_{\mu\nu}\} + K^\alpha{}_{\mu\nu} + L^\alpha{}_{\mu\nu}$$

Here we assume **metric-compatibility**:

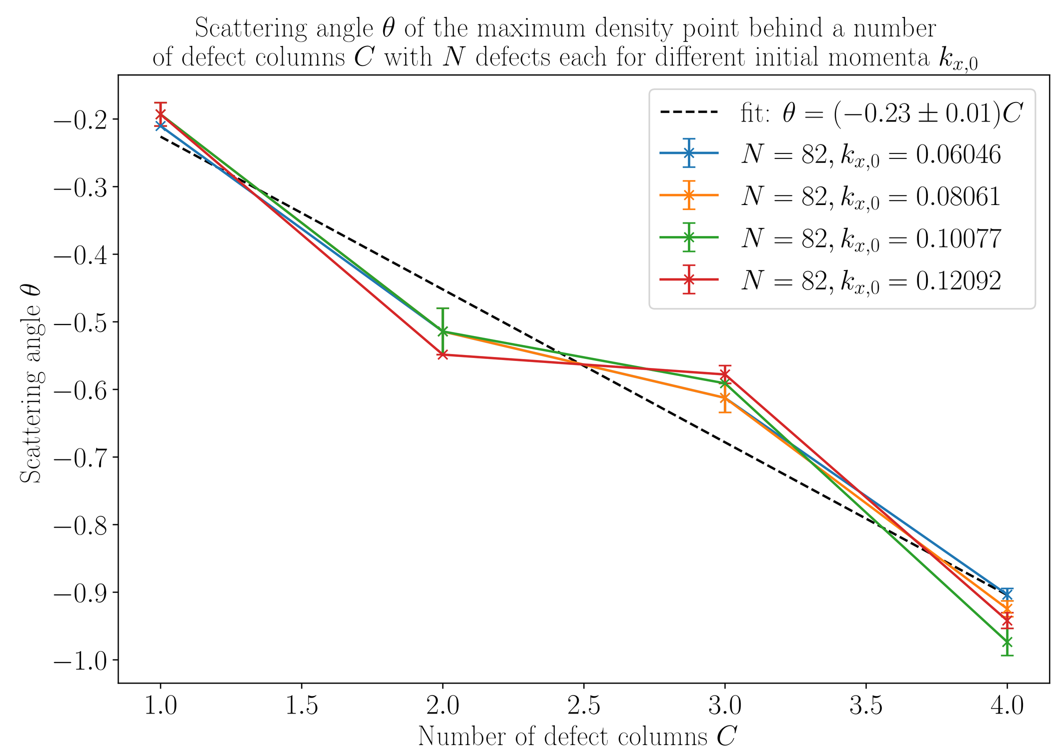
$$\nabla_\lambda g_{\mu\nu} = 0 \Leftrightarrow L^\alpha{}_{\mu\nu} = 0$$

The **contorsion tensor** is build from the torsion:

$$K^\alpha{}_{\mu\nu} = \frac{1}{2} \left(T^\alpha{}_{\mu\nu} - T_{\mu\nu}{}^\alpha + T_\nu{}^\alpha{}_\mu \right)$$

TODO:
In depth visual explanation of
torsion and how it is analogous
to the burgers vector

TODO:
Comparision of models (tight binding vs. autoparallels)
-derivation of scattering angle in torsion picture
-Plot Scattering angle over defects



TODO:
Outlook
- Dirac equation on curved background with torsion
- Numeric simulations of this for some torsion fields

TODO:
References